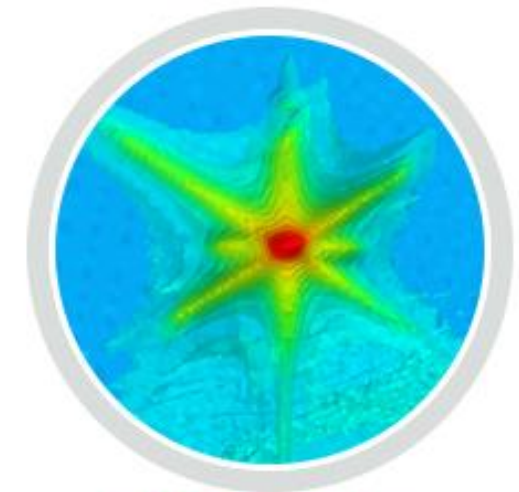
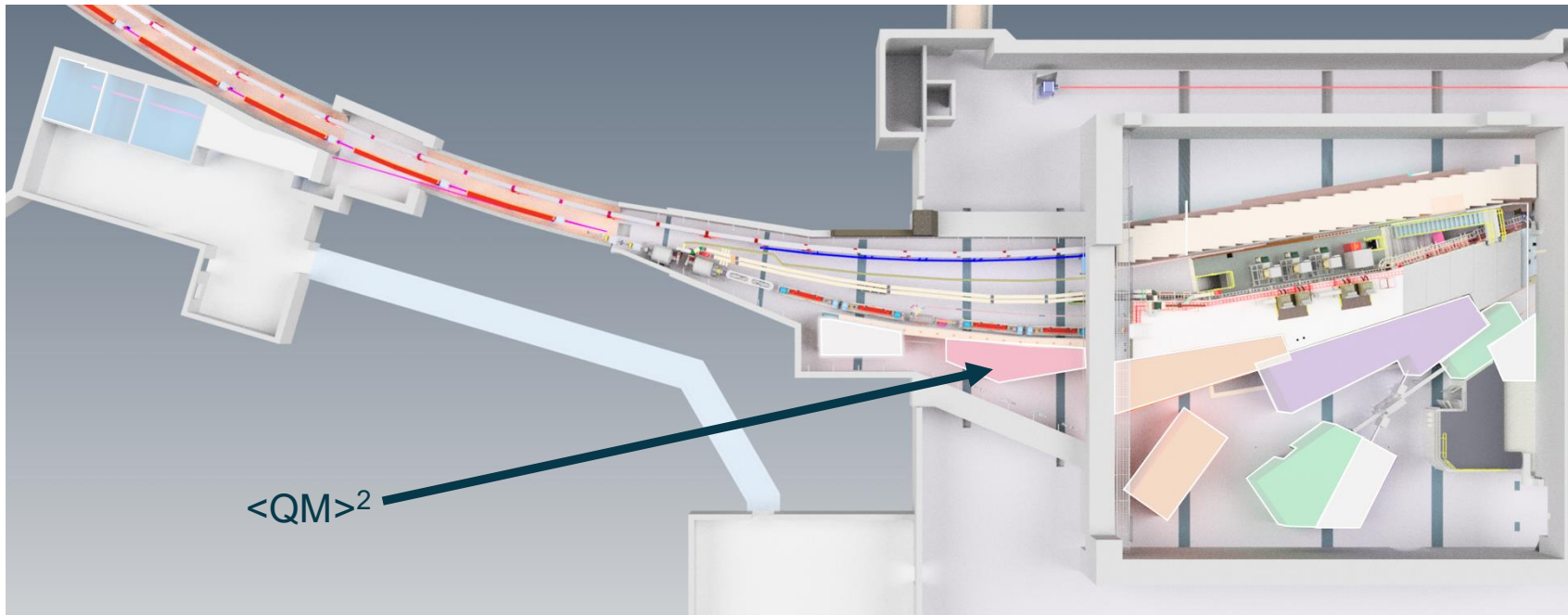


Q-Mapping for Quantum Materials $\langle QM \rangle^2$ Beamline; ID4B

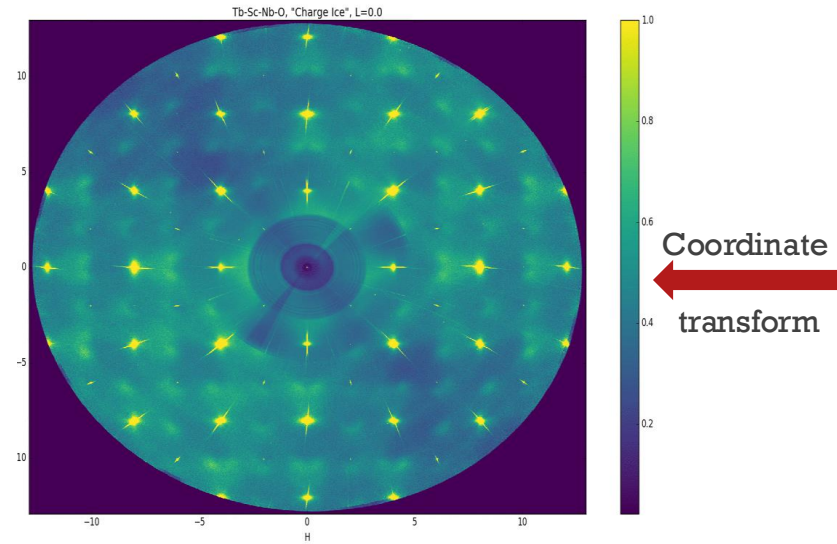


QM2 Beamline

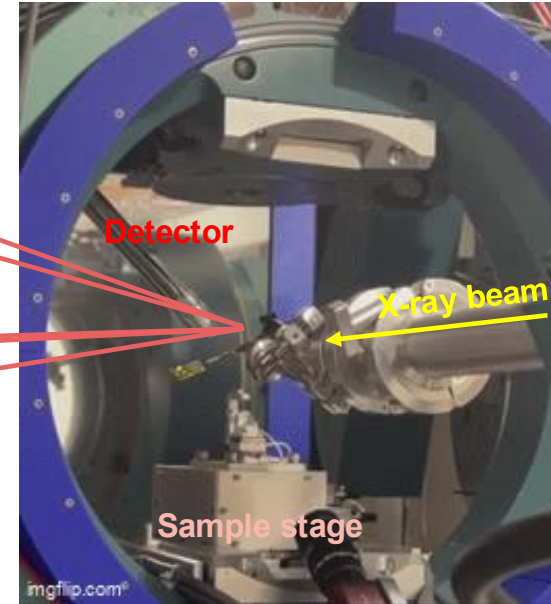
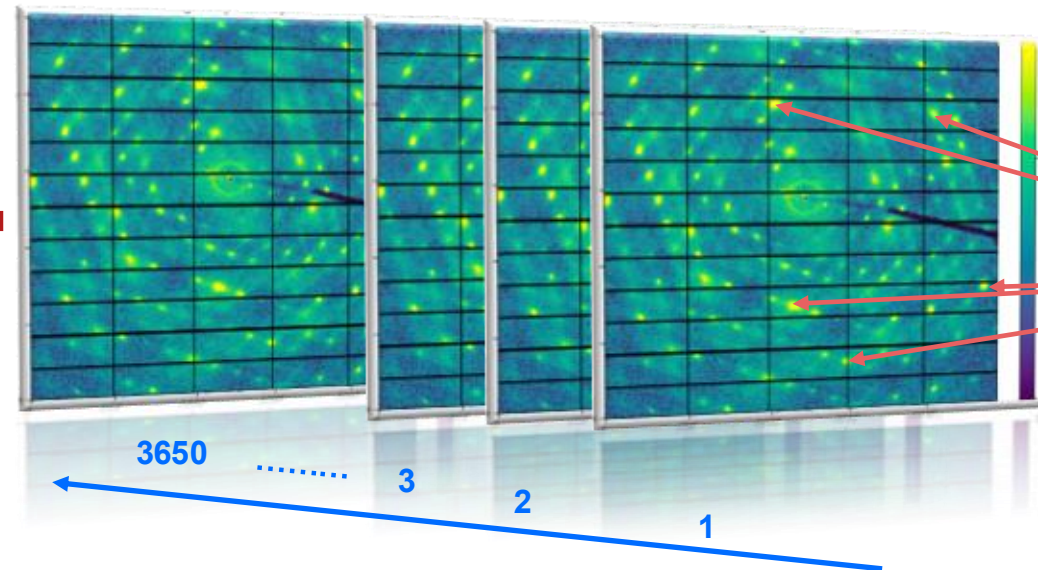
High-throughput characterization of quantum materials.

Data @QM2

Processed data



Raw data



Total frame : 10950 images

Data Structure

Data file : <https://github.com/suchismitasarker/CHESS-Summer-School-QM2/blob/main/project/2026/project1-B0/data/file>

Example of file path : /nfs/chess/id4b/2025-2/he-4025-d/raw6M/Ta2NiSeS70/S1/

```
cd /nfs/chess/id4b/2025-2/he-4025-d/raw6M/Ta2NiSeS70/S1/
```

```
0 drwxrwsr-x 5 chess_id4b root 88 May 11 2025 240
0 drwxrwsr-x 5 chess_id4b root 88 May 11 2025 17
0 drwxrwsr-x 5 chess_id4b root 88 May 11 2025 21
0 drwxrwsr-x 5 chess_id4b root 88 May 11 2025 25
0 drwxrwsr-x 5 chess_id4b root 88 May 11 2025 30
0 drwxrwsr-x 5 chess_id4b root 88 May 11 2025 35
0 drwxrwsr-x 5 chess_id4b root 88 May 11 2025 40
0 drwxrwsr-x 5 chess_id4b root 88 May 11 2025 50
0 drwxrwsr-x 5 chess_id4b root 88 May 11 2025 60
0 drwxrwsr-x 5 chess_id4b root 88 May 11 2025 70
0 drwxrwsr-x 5 chess_id4b root 88 May 11 2025 90
0 drwxrwsr-x 5 chess_id4b root 88 May 11 2025 110
0 drwxrwsr-x 5 chess_id4b root 88 May 12 2025 115
0 drwxrwsr-x 5 chess_id4b root 88 May 12 2025 117
0 drwxrwsr-x 5 chess_id4b root 88 May 12 2025 119
```

```
Ta2NiSeS70_006
Ta2NiSeS70_007
Ta2NiSeS70_008
```

Images

```
Ta2NiSeS70_PIL10_006_00000.cbf
Ta2NiSeS70_PIL10_006_00001.cbf
Ta2NiSeS70_PIL10_006_00002.cbf
Ta2NiSeS70_PIL10_006_00003.cbf
Ta2NiSeS70_PIL10_006_00004.cbf
...
Ta2NiSeS70_PIL10_006_03650.cbf
```

Crystallographic Binary File (CBF)

```
file_path = "/nfs/chess/id4b/2025-2/he-4025-d/raw6M/Ta2NiSeS70/S1/180/Ta2NiSeS70_074/Ta2NiSeS70_PIL10_074_03650.cbf"
```

- The most widely recognized use of .cbf
- Used in scientific research, especially:
 - X-ray crystallography
 - Synchrotron experiments
- Stores binary **image data + metadata** (like detector settings, wavelength, etc.)
- Designed as a compact, efficient version of CIF (Crystallographic Information File)
- Key features:
 - Combines readable headers (ASCII text) with binary data
 - Efficient for large experimental datasets
 - Used by tools like:
 - CBFlib
 - XDS
 - DIALS
 - fabIO

FabIO Package

FabIO is an I/O library for images produced by 2D X-ray detectors and written in Python.

<https://www.silx.org/doc/fabio/latest/api/modules.html>

<https://pypi.org/project/fabio/>

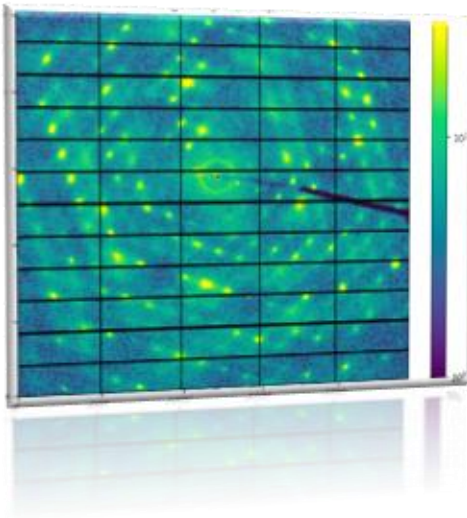
<https://github.com/silx-kit/fabio>

Example: `python read_cbf.py`

https://github.com/suchismitasarker/CHESS-Summer-School_QM2/blob/main/meeting/extra_code/read_cbf.py

Size of the raw data

Raw data



FeTe2_010

File Edit View Go Help

< > ^ Home /nfs/chess/daq/current/id4b/sarker-0000-a/raw6M/FeTe2/distorted/290/FeTe2_010/

Places	Name	Size	Type	Date Modified
Computer	FeTe2_PIL10_010_00000.cbf	6.0 MiB	MATLAB file	Wednesday
chess_id4b	FeTe2_PIL10_010_00001.cbf	5.9 MiB	MATLAB file	Wednesday
Desktop	FeTe2_PIL10_010_00002.cbf	5.9 MiB	MATLAB file	Wednesday
Trash	FeTe2_PIL10_010_00003.cbf	5.9 MiB	MATLAB file	Wednesday
aux/cycles/current	FeTe2_PIL10_010_00004.cbf	5.9 MiB	MATLAB file	Wednesday
daq				
id4b				
id4baux				
Devices				
File System				

Total - 63 GB/temperature

Total frame : 10950 images

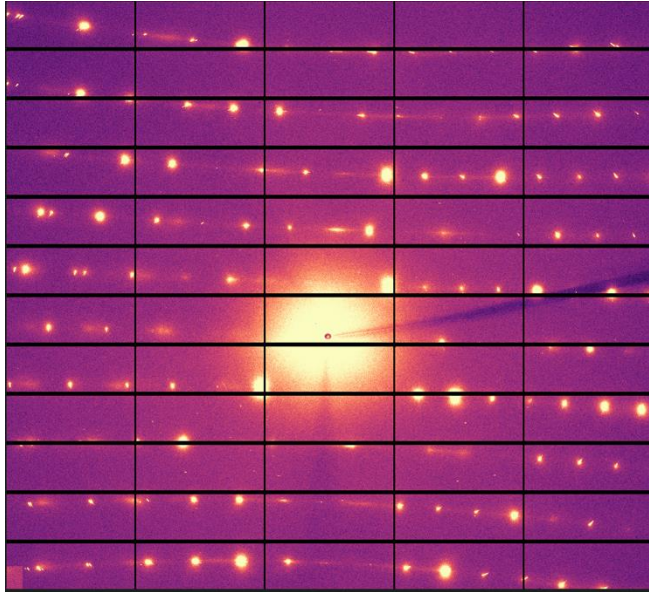
Namachine access:

<https://wiki.classe.cornell.edu/Computing/NoMachine>

Fast Azimuthal Integration using Python : pyFAI

Use the calibration files or mask for the raw data to take care of dead-pixel or shadow

<https://pyfai.readthedocs.io/en/stable/>



```
/nfs/chess/id4baux/2025-2/he-4025-d/calibrations/  
- CeO2.poni  
- mask.edf
```

Data processing software - nexrefine

NeXpy v2.0.0 (on lnx1034-f1.classe.cornell.edu)

File Edit Data Experiment Refine Server View Window Script Help

NeXus Data

VN_16

- entry
 - instrument
 - nxcombine
 - nxreduce
 - sample
 - transform
- f1
 - data
 - instrument
 - logs
 - monitor
 - nxcopy
 - nxfind
 - nxlink
 - nxload
 - nxmax
 - nxrefine
 - nxtransform
 - peaks
 - radial_sum
 - sample
 - scan_number
 - start_time
 - summed_data
 - summed_frames
 - transform
- f2
- f3
- VN_300

Manage Workflows (on lnx1034-f1.classe.cornell.edu)

Choose Sample Directory: /nfs/chess/id4baux/2026-1/de-la-to-4792-a/nxrefine/NaLuSe2/YbLu_2

Choose Parent File: NaLuSe2_15.nxs

Scan	load	link	copy	max	find	refine	prepare	transform	masked	coml
								regular	masked	regular
15	☒	☒	☒	☒	☒	☒	☐	☒	☐	☒
30	☒	☒	☒	☒	☒	☒	☐	☒	☐	☒
50	☒	☒	☒	☒	☒	☒	☐	☒	☐	☒
100	☒	☒	☒	☒	☒	☒	☐	☒	☐	☒
200	☒	☒	☒	☒	☒	☒	☐	☒	☐	☒
300	☒	☒	☒	☒	☒	☒	☐	☒	☐	☒
All	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐

Update Status Add to Queue View Logs

Jupyter QtConsole 5.4.4
 Python 3.11.9 | packaged by conda-forge | (main, Apr 19 2024, 18:36:13) [GCC 12.3.0]
 Type 'copyright', 'credits' or 'license' for more information
 IPython 8.10.0 -- An enhanced Interactive Python. Type '?' for help.
 In [1]:

VN_16:Nxroot @HDF5_Version = '1.14.3'; @creator = 'nexusformat'; @creator_version = '2.0.0'; @default = 'entry'; @file_name = '/nfs/chess/id4baux/2026-1/wils

Processed data

Help

/nfs/chess/id4baux/2025-2/he-4025-d/nxrefine/Ta2NiSeS70/S1/

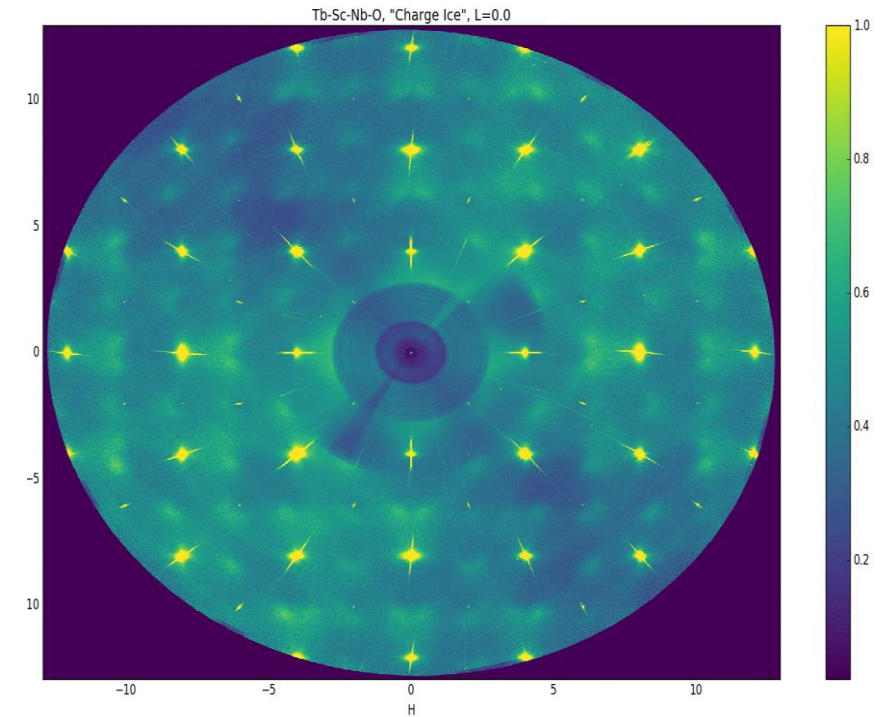
Name	Size	Type	Date Modified
25	10 bytes	folder	05/12/2025
21	10 bytes	folder	05/12/2025
17	10 bytes	folder	05/12/2025
240	10 bytes	folder	05/11/2025
Ta2NiSeS70_160.nxs	95.4 MiB	NoMachine session	05/12/2025
Ta2NiSeS70_150.nxs	95.4 MiB	NoMachine session	05/12/2025
Ta2NiSeS70_180.nxs	95.4 MiB	NoMachine session	05/12/2025
Ta2NiSeS70_140.nxs	95.9 MiB	NoMachine session	05/12/2025
Ta2NiSeS70_170.nxs	95.3 MiB	NoMachine session	05/12/2025
Ta2NiSeS70_129.nxs	96.0 MiB	NoMachine session	05/12/2025

/nfs/chess/id4baux/2026-1/de-la-to-4792-a/nxrefine/NaYbSe2/YbLu_1/140/

Name	Size	Type
f2.h5	8.8 GiB	HDF document
f2_transform.pars	1.6 KiB	plain text document
f1.h5	8.8 GiB	HDF document
f1_transform.pars	1.6 KiB	plain text document
f3.h5	8.8 GiB	HDF document
f3_transform.pars	1.6 KiB	plain text document
f2_transform.nxs	10.6 GiB	NoMachine session
f3_transform.nxs	11.4 GiB	NoMachine session
f1_transform.nxs	10.3 GiB	NoMachine session
transform.nxs	5.8 GiB	NoMachine session

10 files: 64.5 GiB (69,293,154,409 bytes). Free space: 18.3 TiB

Processed



Total - 64.5 GB/temperature

Processed data: NeXus

NeXus is a hierarchical data format built on HDF5.

Structure of a NeXus File

HDF5

Groups (like folders)
 Datasets (like files)
 /entry/data/image → 3D array
 Attributes (metadata)

```
/entry
  /instrument
    /detector
  /sample
  /data
```

<https://nexpy.github.io/nexpy/pythonshell.html>

<https://www.hdfgroup.org/solutions/hdf5/>

Example:

https://github.com/suchismitasarker/CHESS-Summer-School-QM2/blob/main/project/2026/project2-AgenticAI/QM2_data_unwrap.ipynb

https://jupyterhub.classe.cornell.edu/user/ss3428/lab/tree/id4baux_link_jul/nxs_analysis_tools/nxs_analysis_tools-main/docs/source/final/QM2_data_unwrap_summerstudent.ipynb

Different file formats

Feature	HDF5	CBF	NeXus
Purpose	General data storage	Detector images	Standardized scientific data
Structure	Hierarchical	Hybrid (text + binary)	HDF5-based
Flexibility	Very high	Limited	High (but structured)
Use case	Any large dataset	Raw diffraction images	Processed experiment data

Accessing jupyterhub

<https://wiki.classe.cornell.edu/Computing/JupyterHub>

...

#Go to the website : <https://jupyterhub.classe.cornell.edu>

Talk to beamline scientist before selecting the server

#Open the terminal in the jupyterhub

#Go to the specific folder

```
[ss3428@lnx201 ~]$ cd /nfs/chess/sw/qm2_XTEC
```

#Activate the environment

```
[ss3428@lnx201 ~]$ source /nfs/chess/sw/qm2_XTEC/bin/activate
```

#Check the python environment before moving forward

```
[ss3428@lnx201 ~]$ which python
```

#Install ipykernel if needed

```
[ss3428@lnx201 ~]$ pip install ipykernel
```

#It will create environment in the jupyterhub

```
[ss3428@lnx201 ~]$ python -m ipykernel install --user --name=my-python-env --display-name "qm2_XTEC"
```

#Creating link in the jupyterhub folder

```
[ss3428@lnx313 ~]$ ln -s /nfs/chess/id4baux/summer_program ~/summer_program
```

Next week: Task

Tasks: Make small ppt and code - show the progress of the work (Make sure to upload the code in github)

Aditya and Max:

- Python code to import data from multiple folders
 - Each folder contains 3 subfolders
 - Choose the first one
 - Sum of first 10 images
 - Get the statistics
 - Example:
 - Mean intensity comparison
 - Standard deviation comparison
 - Min/Max intensity ranges
 - Total integrated intensity
 - Number of non-zero pixels
 - Median values
 - Intensity ratios (total, peak, mean)
 - Correlation coefficient
 - RMS difference

Olivia : Clustering

- Learn file structure, shapes, trees
- import pipeline Automated data loading of .nxs files